

IP:- UT2 QB.

- 1) What are the features of React Js.
- 2) Give characteristics of RIA.
- 3) Explain the features of PHP and write a PHP program to print factorial of number.
- 4) What is AJAX? Explain its role in web application.
- 5) Differentiate the different data types of PHP.
- 6) Write a code to display "Hello World" in React Js.
- 7) Differentiate between JSON and XML.
- 8) Write a short note on JSX.

IP:- Assignment 2.

1), 2), 3), 4), 5), 6), 7).

15 Oct 2025 *Answer:-

Q. Differentiate betn : JSON and XML.

Parameters	JSON	XML
Abbreviation	It is JavaScript Object Notation.	It is a extensible markup language.
Extended form	It is based on Java Script language.	It is derived from SGML.
Supporting	Supports array	Supports namespace.
Readability	Easy to read	Less easy to read.
Compact code	Less code is created	Requires more code
Speed	Quicker than XML	Slower than JSON.
Syntax	Simple key-value pairs with brackets.	Markup based with custom tags
Size	Smaller than XML	Big size due to markup
Web support	Very limited support for web services.	Huge hold of web services

Q. Write a shortnote on: JSX.

- - JSX stands for JavaScript XML.
- JSX is a language that allows you to mix JavaScript and HTML-like tags to define User Interface (UI) elements and their functionality.
- JSX converts HTML tags into React elements. JSX is a shorthand for calling `React.createElement()` to create elements and components.
- JSX ~~ex~~ looks like HTML and offers a friendly syntax for creating DOM elements.
- Example:

```
const element = <h1>This is Sample JSX </h1>;
```

• The above code snippet somewhat looks like HTML & it also uses a JavaScript like variable but it's neither HTML nor JavaScript, it is JSX.

• JSX is basically a syntax extension of regular JavaScript and is used to create React elements. These elements are then rendered to the React DOM.

- Advantages:

i) It is faster because it performs optimization while compiling code to JavaScript.

ii) It is also type-safe and most of the errors can be caught during compilation.

iii) It makes it easier & faster to write templates.

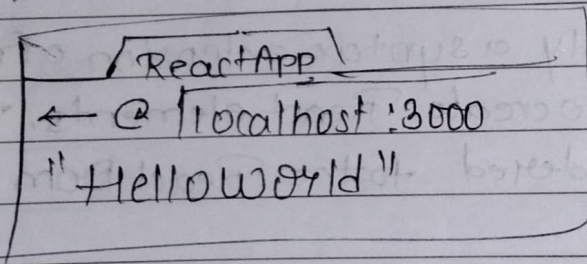
Q) Write a code to display "Hello World" in React Js.

→ [src/App.js]
import React from 'react';
import ReactDOM from 'react-dom';

```
function App() {  
  return (  
    <div>  
      <h1>"Hello World"</h1>  
    </div>  
  );  
}
```

```
export default App;
```

Output:-



• Explanation:-

- We import React & ReactDOM to use React features
- Here, function component is being used & simply Hello World is returned
- After this this component is exported.
- To

• To Run:-

→ To compile the final file run foll. command
npm run start

http://localhost:3000/index.html → URL
open in browser

Q. What are the features of ReactJs.
→ Following are the features of ReactJs.

i) JSX :

It is a JavaScript Syntax extension. JSX stands for JavaScript XML. Although it is not necessary to use JSX, it is recommended to use it.

ii) Components :

Components are the core of React. They are essential when you are handling the code for large-scale project.

iii) One-way data-binding :

React applies a oneway data-binding or a Unidirectional data flow. The advantage of this is that it gives the user better control over the application. Flux ensures the unidirectional flow of data. It increases efficiency by improving the application's flexibility.

iv) Simplicity :

The JSX file in ReactJs simplifies the application.

v) Performer :

React's performance is excellent. The main reason behind it is that it uses a virtual DOM, unlike the other frameworks.

vi) Licence :

ReactJs is licensed under facebook Inc.

vii) Virtual Dom:

React creates a virtual Dom, i.e. in-memory data-structure cache. Only the final changes of Dom have later updated in the browser's Dom.

viii) JavaScript Expressions:

JS expressions can be used in the jsx files using curly brackets, for example `{}`.

Q7 Explain the features of PHP & write the PHP program to print the factorial of a number.

→ • Features of PHP

i) Simple:

It is very simple and easy to use, compared to another scripting language and this is widely used all over the world.

ii) Interpreted:

It is an interpreted language i.e., there is no need for compilation.

iii) Faster:

It is faster than other scripting languages e.g., asp, jsp.

iv) Open Source:

Open Source means you do not need to pay for using PHP, you can free download & use.

v) Platform Independent;

PHP code will run on every platform, linux, Unix, Mac OS X, windows.

vi) Community Support:

large active community good documentation and framework.

• Program:-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head> <title> Loops </title> </head>
```

```
<body> <fontsize=5>
```

```
<?php
```

```
$n=5;
```

```
$f=1;
```

```
while ($n > 0)
```

```
{
```

```
    $f = $f * $n;
```

```
    $n = $n - 1;
```

```
}
```

```
echo "factorial is" . $f;
```

```
?>
```

```
</font>
```

```
</body>
```

```
</html>
```

factorial is 120

5.9.9 PHP Data Types

PHP data types are used to hold different types of data or values. PHP supports 8 primitive data types that can be categorized further in 3 types:

1. Scalar Types (predefined)
2. Compound Types (user-defined)
3. Special Types

PHP Data Types: Scalar Types

It holds only single value. There are 4 scalar data types in PHP.

- | | |
|------------|------------|
| 1. boolean | 2. integer |
| 3. float | 4. string |

PHP Data Types : Compound Types

It can hold multiple values. There are 2 compound data types in PHP.

- | | |
|----------|-----------|
| 1. array | 2. object |
|----------|-----------|

PHP Data Types: Special Types

There are 2 special data types in PHP.

- | | |
|-------------|---------|
| 1. resource | 2. NULL |
|-------------|---------|